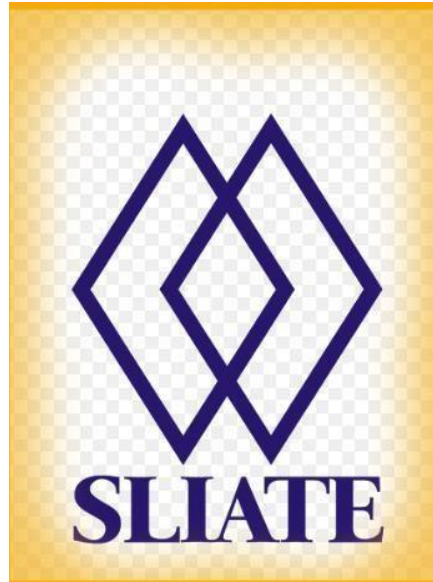


Sri Lanka Institute of Advanced Technological Education



***Higher National Diploma in Information Technology
Advanced Technological institute, Dehiwala***

Database Management System Final report

Name: I.A.Shalini Himajana Fernando

Registration Number: DEH/IT/2324/F/0009

Name: P.A.Tharukshi Hansamali

Registration Number: DEH/IT/2324/F/0068

Hostel Room Allocation & Complaint System

❖ **Real-World Problem Definition**

In many educational institutions, managing hostel rooms and handling student complaints is a complex and time consuming process. Manual systems lead to problems such as improper room allocation, data redundancy, difficulty in tracking student complaints and lack of transparency. To overcome these issues, a computerized hostel room allocation & complaint system is required to efficiently manage hostel details, student room allocations and complaints.

❖ **System Description**

The hostel room allocation & complaint system is designed to manage hostel operations digitally. The system maintains records of hostels, rooms, wardens, students, room allocations and complaints. Student can be allocated rooms for specific time periods and submit complaints related to hostel facilities. Wardens can manage and monitor complaints effectively.

❖ Requirement Analysis

➤ Functional requirements

- The system should store hostel details.
- The system should manage room details and capacity.
- The system should store student information.
- The system should allocate rooms to students.
- The system should allow students to submit complaints.
- The system should track complaint status.

➤ Non-functional requirements

- The system should ensure data consistency.
- The system should reduce data redundancy.
- The system should be secure and reliable.
- The system should be easy to use and maintain.
- The system should provide fast data retrieval.

❖ Entities and Attributes

➤ Hostel

- Hostel_ID (Primary Key)
- Hostel_Name
- Type
- Location

➤ Warden

- Warden_ID (Primary Key)
- Name
- Phone_Number
- Hostel_ID (Foreign Key)

➤ Room

- Room_ID (Primary Key)
- Room_Number
- Capacity
- Hostel_ID (Foreign Key)

➤ Student

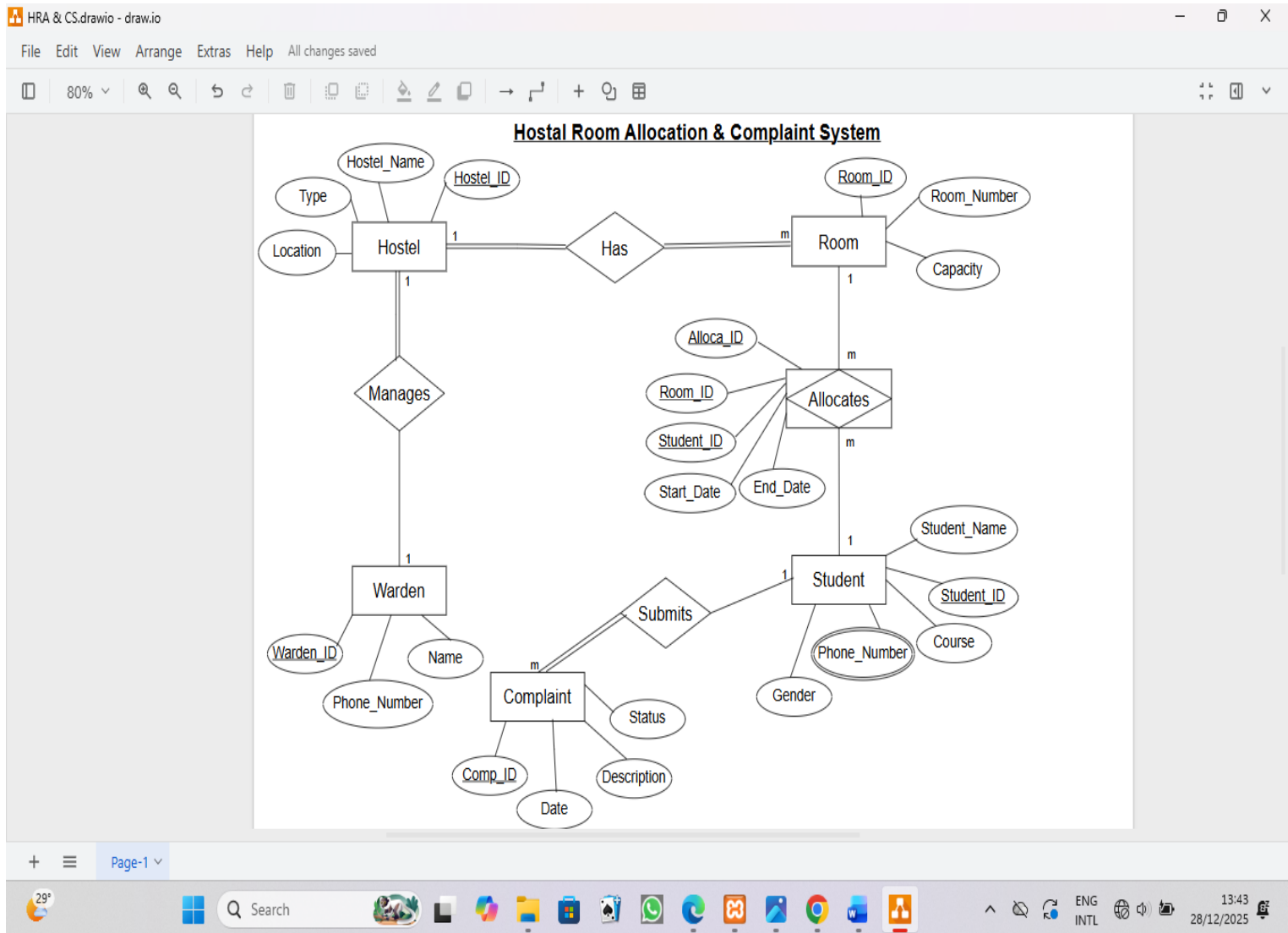
- Student_ID (Primary Key)
- Student_Name
- Gender
- Course
- Phone_Number

- Allocation
 - Alloca_ID (Primary Key)
 - Student_ID (Foreign Key)
 - Room_ID (Foreign Key)
 - Start_Date
 - End_Date
- Complaint
 - Complaint_ID (Primary Key)
 - Description
 - Status
 - Complaint_Date
 - Student_ID (Foreign Key)

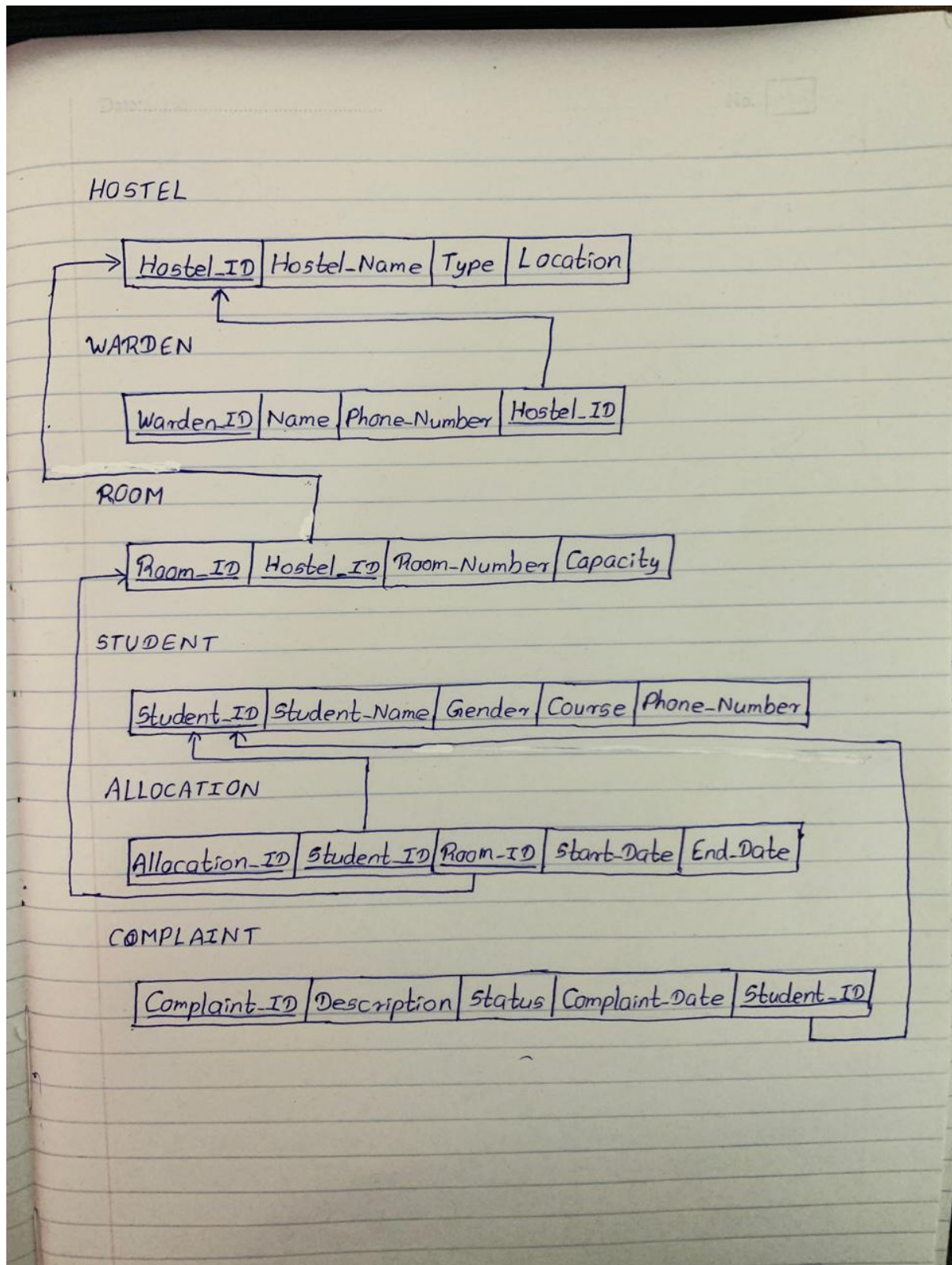
❖ Relationships Between Entities

- One hostel can have many rooms.
- One hostel is managed by one warden.
- One student can have many rooms overtime.
- One room can be allocated to many students overtime.
- One student can submit many complaints.
- Each complaint must be submitted by one student.

❖ ER Diagram



❖ Relational Schema



❖ Database Creation in MYSQL

```
* CREATE DATABASE hostel_management;  
  
USE hostel_management;
```

❖ Table Creation With Keys & Constraints

```
* CREATE TABLE Hostel (  
  
    Hostel_ID INT PRIMARY KEY,  
  
    Hostel_Name VARCHAR(100),  
  
    Type VARCHAR(50),  
  
    Location VARCHAR(100) );  
  
  
* CREATE TABLE Warden (  
  
    Warden_ID INT PRIMARY KEY,  
  
    Name VARCHAR(100),  
  
    Phone_Number VARCHAR(15),  
  
    Hostel_ID INT,  
  
    FOREIGN KEY (Hostel_ID) REFERENCES Hostel(Hostel_ID) );  
  
  
* CREATE TABLE Room (  
  
    Room_ID INT PRIMARY KEY,  
  
    Room_Number VARCHAR(10),  
  
    Capacity INT,  
  
    Hostel_ID INT,  
  
    FOREIGN KEY (Hostel_ID) REFERENCES Hostel(Hostel_ID) );
```



```
* CREATE TABLE Student (  
  
    Student_ID INT PRIMARY KEY,  
  
    Student_Name VARCHAR(100),  
  
    Gender VARCHAR(10),  
  
    Course VARCHAR(100),  
  
    Phone_Number VARCHAR(15) );
```

```
* CREATE TABLE Allocation (  
  
    Allocation_ID INT PRIMARY KEY,  
  
    Student_ID INT,  
  
    Room_ID INT,  
  
    Start_Date DATE,  
  
    End_Date DATE,  
  
    FOREIGN KEY (Student_ID) REFERENCES Student(Student_ID),  
  
    FOREIGN KEY (Room_ID) REFERENCES Room(Room_ID) );
```

```
* CREATE TABLE Complaint (  
  
    Complaint_ID INT PRIMARY KEY,  
  
    Description VARCHAR(255),  
  
    Status VARCHAR(50),  
  
    Complaint_Date DATE,  
  
    Student_ID INT,  
  
    FOREIGN KEY (Student_ID) REFERENCES Student(Student_ID) );
```

❖ SQL Queries

✓ DDL Queries

* ALTER TABLE Student ADD Address VARCHAR(200);
DROP TABLE Complaint;

✓ DML Queries

* INSERT INTO Student VALUES
(01, 'Nimali Silva', 'Female', '0771234567', 'IT');

➔ UPDATE Complaint
SET Status = 'Resolved'
WHERE Complaint_ID = 1;

➔ DELETE FROM Allocation
WHERE Allocation_ID = 1;')

✓ DQL Queries

➔ SELECT * FROM Student;

➔ SELECT * FROM Complaint
WHERE Status = 'Pending';

➔ SELECT * FROM Student
ORDER BY Student_Name;

➔ SELECT Hostel_ID, COUNT(Room_ID) AS Total_Rooms
FROM Room
GROUP BY Hostel_ID
HAVING COUNT(Room_ID) > 5;

❖ Query Outputs

1) Query: `SELECT * FROM Student;`

Answer:

The screenshot shows the phpMyAdmin interface with the 'student' table selected. The query `SELECT * FROM student;` has been executed, resulting in 4 rows of data. The status bar indicates 'Showing rows 0 - 3 (4 total, Query took 0.0005 seconds.)'. The table data is as follows:

Student_ID	Student_Name	Gender	Phone_Number	Course
1	Sunil Perera	Male	0774512358	IT
5	Pawani	Female	0714582480	Accountancy
10	Nimali Silva	Female	0771234567	IT
11	Dahami Fernando	Female	0771212347	Management

2) Query: `SELECT * FROM complaint`

`WHERE Status = 'Pending';`

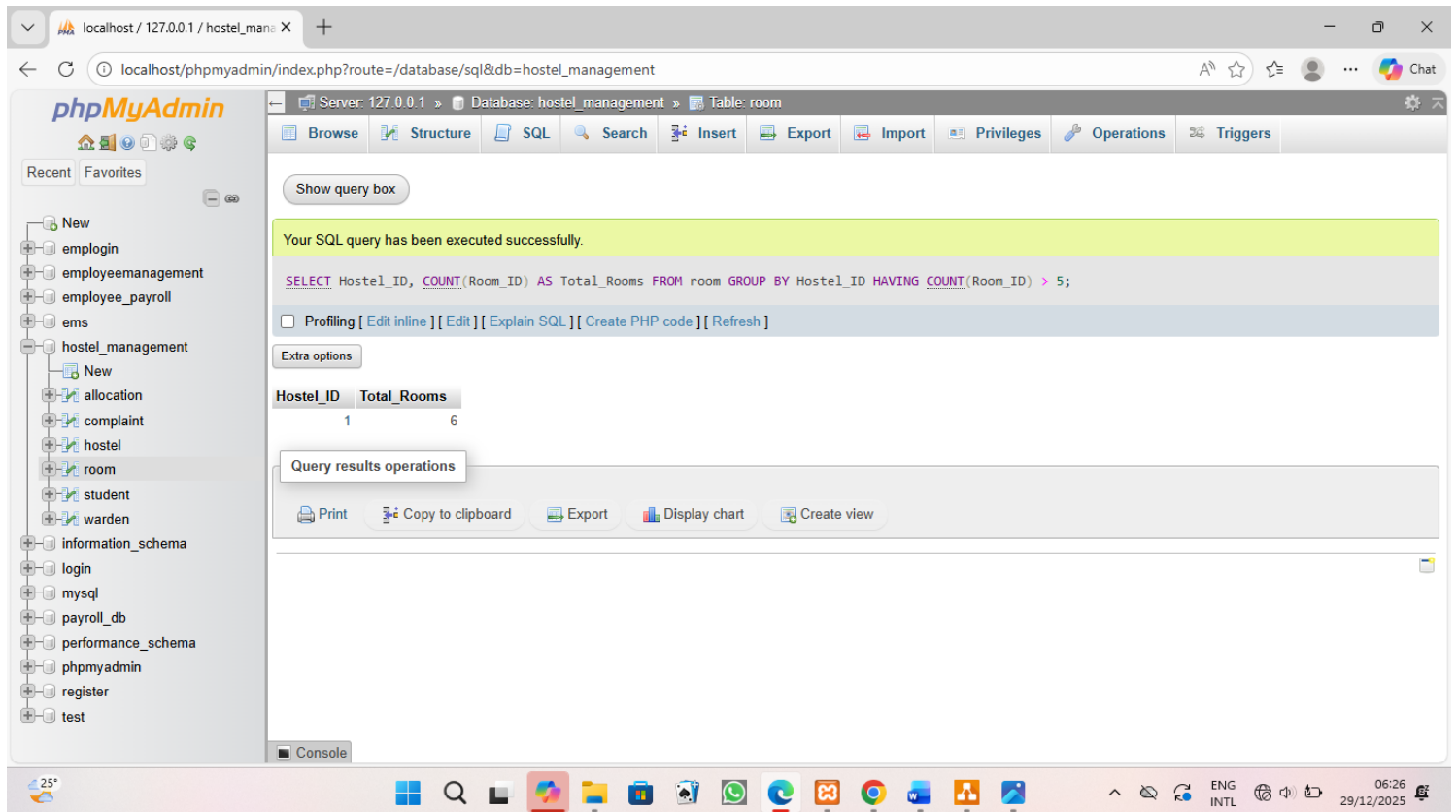
Answer:

The screenshot shows the phpMyAdmin interface with the 'complaint' table selected. The query `SELECT * FROM complaint WHERE Status = 'Pending';` has been executed, resulting in 1 row of data. The status bar indicates 'Showing rows 0 - 0 (1 total, Query took 0.0079 seconds.)'. The table data is as follows:

Complaint_ID	Description	Status	Complaint_Date	student_ID
2	Water leaked	Pending	2025-06-30	5

3) Query: `SELECT Hostel_ID, COUNT(Room_ID) AS
Total_Rooms
FROM room
GROUP BY Hostel_ID
HAVING COUNT(Room_ID) > 5;`

Answer:



The screenshot displays the phpMyAdmin web interface. The left sidebar shows a database structure with 'hostel_management' selected. The main panel shows the 'room' table selected. A yellow message box states 'Your SQL query has been executed successfully.' Below this, the executed query is shown: `SELECT Hostel_ID, COUNT(Room_ID) AS Total_Rooms FROM room GROUP BY Hostel_ID HAVING COUNT(Room_ID) > 5;`. The query results are displayed in a table with two columns: 'Hostel_ID' and 'Total_Rooms'. The results show one row: Hostel_ID 1 with Total_Rooms 6. Below the results, there are buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

Hostel_ID	Total_Rooms
1	6

❖ Conclusion

The hostel room allocation & complaint system efficiently manages hostel operations by reducing redundancy, improving data accuracy and providing better complaint handling.